GGG-C-520b
August 3, 1967
SUPERSEDING
Fed. Spec. GGG-C-520a
June 4, 1964

FEDERAL SPECIFICATION

# CLOTH AND DISKS, ABRASIVE, OPEN MESH, WATERPROOF

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

## 1. SCOPE AND CLASSIFICATION

1.1 Scope. Abrasive cloth covered by this specification is an open mesh or web material uniformly coated on both sides with waterproof resin and laticies, and abrasive grain. This cloth is designed for use by hand or with light-duty portable sanding machines and with floor maintenance machines. As a general rule, aluminum oxide abrasives are used on the harder metals and alloys, and the silicon carbide abrasives are used on softer metals such as aluminum, soft brass, soft bronze, and other materials such as lacquer, glass, and plastics.

1.2 Classification.

1.2.1 Types and classes. Abrasive open mesh cloth covered by this specification shall be of the following types and classes, as specified (see 6.2).

Type I - Aluminum oxide.
- Class 1 - Sheets.
- Class 2 - Rolls.
- Class 3 - Disks.

Type II - Silicon carbide.
- Class 1 - Sheets
- Class 2 - Rolls.
- Class 3 - Disks.

## 2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specification:

CCC-T-191 - Textile, Test Methods.

FSC 5350

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:40Z
Check the source to verify that this is the current version before use.

Federal Standard:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at theprices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Singles copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL-P-116 - Preservation, Methods of.
MIL-P-3816 - Abrasives and Abrasive Products for Shipment and Storage; Packaging and Packing of.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

U. S. Department of Commerce:

PS-8 - Grading of Abrasive Grain on Coated Abrasive Products.

(Application for copies should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C., 20402.)



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:40Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 Material. Materials used in the manufacture of abrasive open mesh cloth shall be as herein specified and shall meet the applicable test requirements.

3.2 Abrasives.

3.2.1 Type I, aluminum oxide. Aluminum oxide abrasive shall be artificial corundum free from extraneous material.

3.2.2 Type II, silicon carbide. Silicon carbide abrasive shall be free from extraneous material.

3.2.3 Grit sizes. Abrasive shall be of the following grit sizes, as specified (see 6.2).

Type I - Aluminum oxide - 400, 360, 320, 280, 240, 220, 180 and 150.
Type II - Silicon carbide - 600, 500, 400, 360, 320, 280, 240, 220, 180, 150, 120, 100, 80 and 60.

3.2.3.1 Grading, control and tolerances. The grading, control and tolerances of the abrasive grain shall be in accordance with PS-8 for waterproof products.

3.3 Backing. Backing shall be woven of good quality rayon or other synthetic fiber thread suitable for the purpose intended. The cloth shall be a plain or leno weave of single or double ply yarns with a nominal thread count in the range of 12 by 12 to 25 by 25 warp and filler yarns per inch. In any of these substantially square woven cloths, the nominal count of warp and filling yarn may differ by two yarns per inch.

3.3.1 Breaking strength. The breaking strength of the finished product shall be not less than 28 pounds per inch in the direction of the warp and 18 pounds in the direction of the fill.

3.3.2 Backing thread distortion. The warp thread of the finished product shall be parallel to the edge within two threads in 11 inches on sheets and rolls.

3.4 Adhesive. Adhesive material shall be waterproof adhesives including synthetic heat hardening resins and laticies suitable for the purpose intended.

3.5 Coating. The coating shall be of a closed coat type and shall completely cover the exposed surface of the backing with only occasional scattered mesh openings completely closed.

3.6 Identification marking. Each unit package of abrasive sheets, disks and rolls shall be marked in a plain and legible manner with the grit size,



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:40Z
Check the source to verify that this is the current version before use.

the abrasive type or manufacturer's trade designation thereof and the manufacturer's name or trade mark of such known character that the source of manufacture may be readily determined. In addition, class 2 rolls shall each be marked in the same manner and the coated abrasive product itself shall also be printed in the same manner. The pattern of printing the coated abrasive product shall be such that each individual sheet or disk shall show the same information as the marking of unit packages and each roll shall show a portion of such printing.

3.7 Forms and dimensions. Open mesh abrasive cloth shall be of the forms and dimensions shown in table I, as specified (see 6.2).

TABLE I. Forms and dimensions

| Forms1/ | Dimensions | Tolerances |
|---|---|---|
| Class 1, sheets | 9 x 11 inches<br>3-2/3 x 9 inches<br>3 x 8 inches | All sheets ± 1/8 inch length, ± 1/16 inch width |
| Class 2, rolls | 1, 1-1/2, or 2 inch width by 25 yards length2/ | All rolls ± 1/16 inch width, + 12, -0 inch in length |
| Class 3, disks (without center holes), unless otherwise specified (see 6.2)3/ | 5, 6, 7, 8, 9, 10 inch outside diameter | All ± 1/16 inch outside diameter. |
| | 12, 13, 14, 15, 16, 17, 18 inch outside diameter | All disks ± 1/8 inch outside diameter. |
| | 19, 20, 21, 22, 23, 24 inch outside diameter | All disks ± 1/4 inch outside diameter. |

1/The popular sizes only of sheets, rolls, and discs are listed, however, other sizes are available and, if required, should be specified (see 6.2).
2/Up to 2 breaks per roll will be permitted.
3/If center holes are desired in class 3 disks, specify size required (see 6.2).



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:40Z
Check the source to verify that this is the current version before use.

3.7.1 Dimensional tolerances. Tolerance on the physical size of sheets, rolls and disks shall be as specified in table I under standard conditions of 70°F., plus or minus 2°F. and 50 percent relative humidity plus or minus 2 percent.

3.8 Workmanship shall be of the highest grade throughout and shall be equal in every respect to good commercial practice. The abrasive cloth shall lie flat when placed on a plane surface and shall conform to the requirements of 3.3.2 with respect to distortion. It shall also be free of any other imperfections which may adversely affect its durability or serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling.

4.2.1 Inspection lot. All abrasive cloth of the same type, class, grit size and dimensions offered for delivery at one time shall be considered a lot for purposes of sampling and inspection.

4.2.2 Sampling of sheet and disks. Sample packages of sheets and disks shall be selected in accordance with MIL-STD-105 at inspection level I for the examination specified in 4.3. The acceptable quality level shall be 1.5 percent defective. All pieces in the sample packages shall be inspected.

4.2.3 Sampling or rolls. Sample rolls shall be selected in accordance with inspection level II of MIL-STD-105 for the examination of 4.3. At least 8 yards of each sample roll shall be inspected.

4.2.4 Sampling for strength test. Sample sheets and disks shall be selected in accordance with inspection level S-2; not more than one piece from any package. Samples from rolls, shall be selected by counting each roll as 300 pieces, and the sample size shall then be in accordance with inspection level S-2; not more than one test piece shall be selected from any one roll.

4.3 Examination. Each of the samples selected in accordance with 4.2.2 shall be visually and dimensionally examined for compliance with respect to material, workmanship, dimensions distortion, warpage, marking, and other requirements of this specification which may be determined by visual examination.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:40Z
Check the source to verify that this is the current version before use.

4.4 Test procedures.

4.4.1 Breaking strength test. The breaking strength shall be determined by cut strip in accordance with method 5102 of CCC-T-191, except that the jaws shall be set 5 inches apart and six specimens shall be cut both longitudinally (direction of warp) and transversely (direction of fill). Samples tearing at the jaws of the tensile test machine shall be discarded. The specimens shall be conditioned for 24 hours immediately prior to the test in an atmosphere maintained at a temperature of 70° ± 2°F., and at 50 ± 2 percent relative humidity (see 3.3.1).

4.5 Examination of preparation for delivery. The packaging, packing, and marking of the cloth and disks shall be examined to determine compliance with the requirements of section 5.

4.5.1 Examination of packaging requirements not covered by referenced specification shall be performed in accordance with the examination requirements of MIL-P-116.

4.5.2 Examination of packaging and marking requirements not covered by referenced specifications shall be in accordance with MIL-STD-105, using an AQL of 4.0 percent defective.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C, as specified (see 6.2).

5.1.1 Level A. Material shall be packaged in accordance with level A of MIL-P-3816.

5.1.2 Level C. Material shall be packaged in accordance with the suppliers commercial practice.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. Material packaged as specified in 5.1 shall be packed in accordance with level A of MIL-P-3816.

5.2.2 Level B. Material packaged as specified in 5.1 shall be packed in accordance with level B of MIL-P-3816.

5.2.3 Level C. Material packaged as specified in 5.1 shall be packed in accordance with the suppliers commercial practice to assure acceptance by carrier and safe delivery to destination. Containers shall comply with carrier rules and regulations applicable to the mode of transportation.

5.3 Marking.

5.3.1 Civil agencies. In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with Fed. Std. No. 123.



Source: https://assist.dla.mil -- Downloaded: 202610-26T12:40Z
Check the source to verify that this is the current version before use.

5.3.2 Military agencies. In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Abrasive open mesh cloth is generally used for abrading and smoothing wood, metal, cork, fiberglass, prime paint coats, for floor maintenance and most any other applications where loading and filling presents a problem. This cloth in sheets, and disks 10 inches or less in diameter may be used either wet or dry, either by hand or with light duty portable sanders. Disks having a diameter of 12 inches or larger are for use with floor maintenance.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type, class, dimensions, and grit sizes of abrasive open mesh cloth required.
(c) If disk, specify center hole diameter desired, otherwise disks will be supplied without center hole.
(d) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

MILITARY CUSTODIANS:
Army - MR
Navy - SH
Air Force - 84

Review activities:
Army - MR
Navy - SH
Air Force - 84, 85

User activities:
Army - AT, MD, MU
Navy - MC, YD

Preparing activity:
Navy - SH

---

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain copies of documents referenced herein. Price 6 cents each.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:40Z
Check the source to verify that this is the current version before use.